Implementation of Bank Functionality Using Dynamic Polymorphism

■ Objective / Aim

To write a Java program that implements basic bank functionality using classes such as Customer, Acc, RBI, and derived bank classes such as SBI, ICICI, and PNB, and to demonstrate dynamic polymorphism (runtime polymorphism) through method overriding.

■■ System Requirements

Hardware	Standard PC setup (Core i3/i5 Processor, 4GB/8GB RAM)
Operating System	Windows / Linux / macOS
Software	Java Development Kit (JDK 17 or JDK 21)
IDE / Editor	Eclipse / IntelliJ IDEA / NetBeans / VS Code / Text Editor

■■ Algorithm / Logic Description

1. Create a Customer class containing customer ID and name, with a method to display customer details.

2. Create an Acc class containing account number and balance, with methods for deposit, withdrawal, and balance display.

3. Create RBI as the base class and define getRateOfInterest() returning the minimum RBI rate of 4%.

4. Create SBI, ICICI, and PNB classes by extending RBI.

5. Override getRateOfInterest() in each derived bank class with the bank's applicable rate.

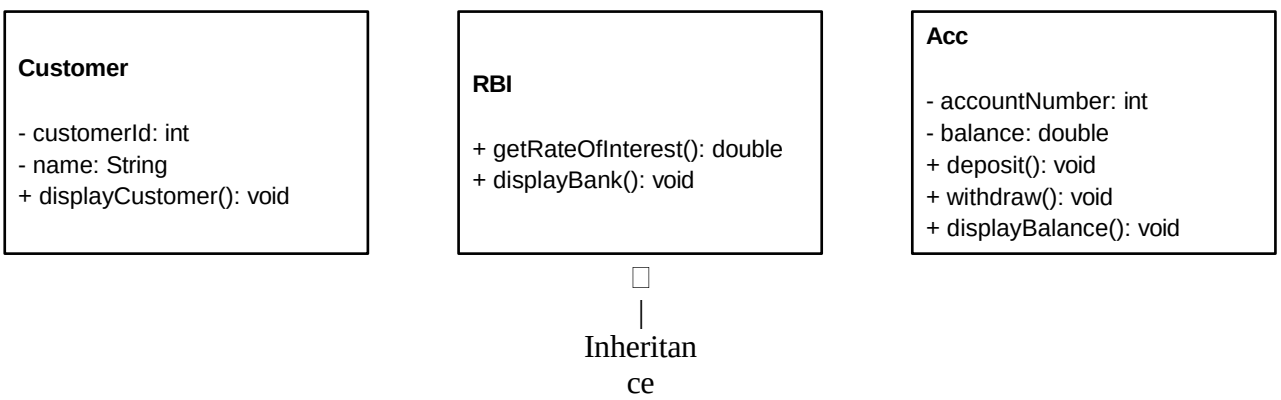
6. In main(), read the bank name using Scanner.

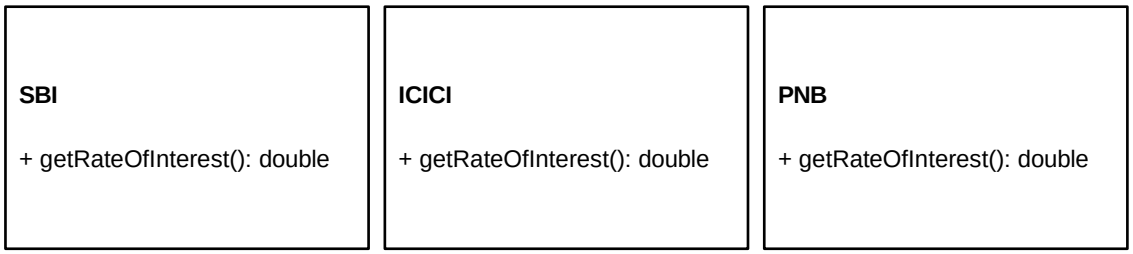
7. Declare an RBI reference variable and assign it an RBI, SBI, ICICI, or PNB object according to the input.

8. Call displayBank(). Because getRateOfInterest() is overridden, Java selects the appropriate implementation at runtime.

9. This runtime method selection demonstrates dynamic polymorphism.

■ UML Class Diagram





Relationship: SBI, ICICI, and PNB extend RBI. The RBI reference variable can refer to an object of any of these classes, allowing runtime method overriding.

■ Source Code

```
package Excersice;

import java.util.Scanner;

// Customer class class
Customer {
	int customerId;
	String name;

	Customer(int customerId, String name) { this.customerId
		= customerId; this.name = name;
	}

	void displayCustomer() {
		System.out.println("Customer ID                    : " + customerId);
		System.out.println("Customer Name : " + name);
	}
}

// Account class class Acc
{
	int accountNumber; double
	balance;

	Acc(int accountNumber, double balance) {
		this.accountNumber = accountNumber; this.balance
		= balance;
	}

	void deposit(double amount) { if (amount
		> 0) {
			balance += amount; System.out.println("Deposited : " +
			amount);
		}
	}

	void withdraw(double amount) {
		if (amount > 0 && amount <= balance) { balance -= amount;
			System.out.println("Withdrawn : " + amount);
		} else {
			System.out.println("Insufficient  balance");
		}
	}

	void displayBalance() { System.out.println("Balance : " + balance);
	}
}

// RBI is the base class class RBI {
	double getRateOfInterest() { return 4.0;
	}

	void displayBank() {
		System.out.println("RBI rate of interest is : "
				+ getRateOfInterest() + "%");
	}
}

// SBI derived class class SBI
extends RBI {
	@Override
	double getRateOfInterest() { return 7.0;
	}
}

// ICICI derived class class ICICI
extends RBI {
	@Override
	double getRateOfInterest() { return 6.5;
	}
```

```
}

// PNB derived class class PNB
extends RBI {
	@Override
	double getRateOfInterest() { return 6.0;
	}
}

// Main class
public class BankROI {
	public static void main(String[] args) { Scanner sc = new
		Scanner(System.in);

		System.out.print(
			"Enter the Bank name to find the rate of Interest : "); String bankName =

		sc.nextLine();

		// RBI reference variable RBI rbi;

		if (bankName.equalsIgnoreCase("RBI")) { rbi = new
			RBI();
		}
		else if (bankName.equalsIgnoreCase("SBI")) { rbi = new
			SBI();
		}
		else if (bankName.equalsIgnoreCase("ICICI")) { rbi = new ICICI();
		}
		else if (bankName.equalsIgnoreCase("PNB")) { rbi = new
			PNB();
		}
		else {
			System.out.println("Invalid bank name"); sc.close();
			return;
		}

		// Runtime polymorphism rbi.displayBank();

		sc.close();
	}
}
```

■ Compilation & Execution Commands

Save the file as:
BankROI.java

Compile:
javac -d . BankROI.java

Run:
java Excersice.BankROI

■■ Output Screen

Testcase 1

```
Enter the Bank name to find the rate of Interest : RBI
RBI rate of interest is : 4.0%
```

Testcase 2

```
Enter the Bank name to find the rate of Interest : SBI
RBI rate of interest is : 7.0%
```

Note: The output text is based directly on the supplied program. Since the code returns double values (4.0 and 7.0), the console displays 4.0% and 7.0%.